

Additional file 1: Supplementary tables

No Single Architecture Wins: Ensembles and Conventional Machine Learning
Match Pretrained Molecular Models Across 45 Property-Prediction Benchmarks

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This file contains the five supplementary tables referenced in the main text. All are generated directly from the deposited benchmark artifacts by `portable_colab_qsar_bundle/render_manuscript_assets.py` and `render_latex_tables.py`, and carry the same values as the machine-readable `manuscript_assets/tables/*.csv` files in the repository.

All results derive from a single split and a single seed per dataset; as set out in the Limitations section of the main text, individual entries are therefore provisional and close comparisons between models should not be treated as established.

Table S1: Per-dataset winning model under the ranking metric, with the corresponding cross-validation-selected model and its relative gap to the dataset best (%).

dataset	suite	task_ kind	family	model	analysis_ metric	analysis_ metric_value	cv_ selected_ model	cv_ selected_ family	cv_selected_gap_to_best
tdc_ames	TDC	classification	Ensemble (stacking / averaging)	Ensemble (OOF Stacking (RidgeCV, 5- fold))	test_ roc_ auc	0.876	Extra trees	Conventional ML	0.577
tdc_bbb_ martins	TDC	classification	Conventional ML	Random forest	test_ roc_ auc	0.925	Tabular MLP	Conventional ML	2.943
tdc_ bioavailability_ ma	TDC	classification	Conventional ML	CatBoost	test_ roc_ auc	0.777	Logistic Regression	Conventional ML	7.943
tdc_ carcinogens_ lagunin	TDC	classification	Chemprop v2 GNN	Chemprop v2 (AttentiveFP, ensemble=3)	test_ roc_ auc	0.929	Voting Classifier (KNN, SVM)	Conventional ML	9.918
tdc_clintox	TDC	classification	CFA combinatorial fusion	CFA (Combinatorial Fusion)	test_ roc_ auc	0.974	Logistic Regression	Conventional ML	8.005
tdc_cyp1a2_ veith	TDC	classification	Ensemble (stacking / averaging)	Ensemble (Weighted average (inverse train RMSE))	test_ roc_ auc	0.971	ChemML MLP (Tensor Flow)	Deep tabular NN (ChemML MLP)	1.432
tdc_cyp2c19_ veith	TDC	classification	Ensemble (stacking / averaging)	Ensemble (Weighted average (inverse train RMSE))	test_ roc_ auc	0.923	XGBoost	Conventional ML	8.976
tdc_cyp2c9_ substrate_ carbonmangels	TDC	classification	Ensemble (stacking / averaging)	Ensemble (OOF Stacking (RidgeCV, 5- fold))	test_ auc	0.466	SVC	Conventional ML	35.513
tdc_cyp2c9_ veith	TDC	classification	Ensemble (stacking / averaging)	Ensemble (OOF Stacking (RidgeCV, 5- fold))	test_ auprc	0.814	AdaBoost	Conventional ML	13.326
tdc_cyp2d6_ substrate_ carbonmangels	TDC	classification	Uni-Mol V1 (3D pretrained)	Uni-Mol V1	test_ auprc	0.652	AdaBoost	Conventional ML	4.966
tdc_cyp2d6_ veith	TDC	classification	Ensemble (stacking / averaging)	Ensemble (OOF Stacking (RidgeCV, 5- fold))	test_ auprc	0.734	AdaBoost	Conventional ML	16.154
tdc_cyp3a4_ substrate_ carbonmangels	TDC	classification	Conventional ML	Random forest	test_ auprc	0.706	Logistic Regression	Conventional ML	2.791
tdc_cyp3a4_ veith	TDC	classification	Ensemble (stacking / averaging)	Ensemble (OOF Stacking (RidgeCV, 5- fold))	test_ auprc	0.884	AdaBoost	Conventional ML	9.750

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dataset	suite	task_ kind	family	model	analysis_ metric	analysis_ metric_value	cv_ selected_ model	cv_ selected_ family	cv_selected_gap_to_best
tdc_dili	TDC	classification	Uni-Mol V1 (3D pretrained)	Uni-Mol V1	test_ roc_ auc	0.922	Logistic Regression	Conventional ML	19.151
tdc_herg	TDC	classification	Conventional ML	AdaBoost	test_ roc_ auc	0.848	Logistic Regression	Conventional ML	16.386
tdc_herg_ karim	TDC	classification	Ensemble (stacking / averaging)	Ensemble (Weighted average (inverse train RMSE))	test_ roc_ auc	0.905	ChemML MLP (Py Torch)	Deep tabular NN (ChemML MLP)	3.672
tdc_hia_hou	TDC	classification	Ensemble (stacking / averaging)	Ensemble (OOF Stacking (RidgeCV, 5- fold))	test_ roc_ auc	0.988	Logistic Regression	Conventional ML	0.500
tdc_pampa_ ncats	TDC	classification	Uni-Mol V1 (3D pretrained)	Uni-Mol V1	test_ roc_ auc	0.763	Logistic Regression	Conventional ML	26.281
tdc_pgp_ broccatelli	TDC	classification	Uni-Mol V1 (3D pretrained)	Uni-Mol V2 (84m)	test_ roc_ auc	0.933	Logistic Regression	Conventional ML	6.966
tdc_skin_ reaction	TDC	classification	Uni-Mol V1 (3D pretrained)	Uni-Mol V2 (84m)	test_ roc_ auc	0.658	Voting Classifier (KNN, SVM)	Conventional ML	23.497
tdc_tox21	TDC	classification	Conventional ML	AdaBoost	test_ roc_ auc	0.556	SVC	Conventional ML	15.985
tdc_toxcast	TDC	classification	Conventional ML	HistGradientBoosting	test_ roc_ auc	0.716	Logistic Regression	Conventional ML	16.900
chemml_cep_ homo	ChemML	regression	Ensemble (stacking / averaging)	Ensemble (Weighted average (inverse train RMSE))	test_ rmse	0.099	Elastic NetCV	Conventional ML	22.270
chemml_ organic_density	ChemML	regression	Chemprop v2 GNN	Chemprop v2 (D- MPNN + RDKit2D, ensemble=3)	test_ rmse	0.005	Elastic NetCV	Conventional ML	34.525
esol_delaney	Molecular Net	regression	Ensemble (stacking / averaging)	Ensemble (Weighted average (inverse train RMSE))	test_ rmse	0.617	Elastic NetCV	Conventional ML	17.596
freesolv_sampl	Molecular Net	regression	Deep tabular NN (ChemML MLP)	ChemML MLP (Py Torch)	test_ rmse	0.993	CatBoost	Conventional ML	115.432
lipophilicity	Molecular Net	regression	Ensemble (stacking / averaging)	Ensemble (Weighted average (inverse train RMSE))	test_ rmse	0.589	Elastic NetCV	Conventional ML	15.769
poduam_pod_ nc_std	PODUAM	regression	Conventional ML	Random forest	test_ rmse	0.720	Elastic NetCV	Conventional ML	9.929

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dataset	suite	task_ kind	family	model	analysis_ metric	analysis_ metric_ value	cv_ selected_ model	cv_ selected_ family	cv_selected_gap_to_best
poduam_pod_ rd_std	PODUAM	regression	Conventional ML	Random forest	test_ rmse	0.572	Elastic NetCV	Conventional ML	22.906
polaris_adme_ fang_hppb_1	Polaris	regression	MapLight + GNN	MapLight + GNN (Cat Boost, Strict Parity)	test_ rmse	0.448	Elastic NetCV	Conventional ML	22.177
polaris_adme_ fang_perm_1	Polaris	regression	Uni-Mol V1 (3D pretrained)	Uni-Mol V1	test_ rmse	0.399	ChemML MLP (Py Torch)	Deep tabular NN (ChemML MLP)	15.413
polaris_adme_ fang_rclint_1	Polaris	regression	Uni-Mol V1 (3D pretrained)	Uni-Mol V1	test_ rmse	0.522	Elastic NetCV	Conventional ML	15.664
polaris_adme_ fang_rppb_1	Polaris	regression	MapLight + GNN	MapLight + GNN (Cat Boost, Strict Parity)	test_ rmse	0.493	Elastic NetCV	Conventional ML	21.893
polaris_adme_ fang_solu_1	Polaris	regression	Uni-Mol V1 (3D pretrained)	Uni-Mol V2 (84m)	test_ rmse	0.551	ChemML MLP (Py Torch)	Deep tabular NN (ChemML MLP)	16.215
tdc_caco2_ wang	TDC	regression	Ensemble (stacking / averaging)	Ensemble (Weighted average (inverse train RMSE))	test_ rmse	0.342	Elastic NetCV	Conventional ML	22.979
tdc_clearance_ hepatocyte_az	TDC	regression	Ensemble (stacking / averaging)	Ensemble (Weighted average (inverse train RMSE))	test_ rmse	43.841	CatBoost	Conventional ML	12.399
tdc_clearance_ microsome_az	TDC	regression	Uni-Mol V1 (3D pretrained)	Uni-Mol V2 (84m)	test_ rmse	33.507	XGBoost	Conventional ML	16.477
tdc_half_life_ obach	TDC	regression	Ensemble (stacking / averaging)	Ensemble (Weighted average (inverse train RMSE))	test_ rmse	18.541	Random forest	Conventional ML	21.086
tdc_ hydrationfreeenergy_- freesolv	TDC	regression	Ensemble (stacking / averaging)	Ensemble (OOF Stacking (RidgeCV, 5- fold))	test_ rmse	1.112	Elastic NetCV	Conventional ML	18.818
tdc_ld50_zhu	TDC	regression	Deep tabular NN (ChemML MLP)	ChemML MLP (Tensor Flow)	test_ rmse	0.828	XGBoost	Conventional ML	2.079
tdc_ lipophilicity_ astrazeneca	TDC	regression	Uni-Mol V1 (3D pretrained)	Uni-Mol V1	test_ rmse	0.617	ChemML MLP (Py Torch)	Deep tabular NN (ChemML MLP)	17.583
tdc_ppbr_az	TDC	regression	MapLight + GNN	MapLight + GNN (Cat Boost, Strict Parity)	test_ rmse	11.356	ChemML MLP (Py Torch)	Deep tabular NN (ChemML MLP)	27.931
tdc_solubility_ aqsolddb	TDC	regression	Uni-Mol V1 (3D pretrained)	Uni-Mol V1	test_ rmse	1.007	XGBoost	Conventional ML	1.814

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dataset	suite	task_ kind	family	model	analysis_ metric	analysis_ metric_value	cv_ selected_ model	cv_ selected_ family	cv_selected_gap_to_best
tdc_vdss_ lombardo	TDC	regression	MapLight + GNN	MapLight + GNN (Cat Boost, Strict Parity)	test_ rmse	4.670	Extra trees	Conventional ML	50.704

Table S2: Staged component ablation: datasets whose best achievable primary metric improved when each pipeline stage was added, computed from the existing model results.

stage_order	stage	datasets_evaluated	datasets_improved_vs_previous	improvement_fraction
1	Conventional ML only	44	0	0.000
2	+ Map Light classic fea- tures	44	10	0.227
3	+ neu- ral/ deep back- ends	44	26	0.591
4	+ CFA fusion	44	6	0.136
5	Full pipeline incl. ensem- bles	44	16	0.364

Table S3: Feature-family selection summary across all datasets, sorted by per-feature enrichment relative to a uniform-selection baseline.

Feature family	Datasets selected	Available (sum)	Selected (sum)	% of selected	Enrichment vs uniform
rdkit	40	8,261	1,362	11.22	5.16
erg	38	12,552	1,146	9.44	2.86
avalon	43	44,384	3,603	29.69	2.54
maccs	38	5,962	224	1.85	1.18
maplight	41	30,068	854	7.04	0.89
fcfp6	43	42,500	998	8.22	0.73
atom_pair	41	39,651	830	6.84	0.66
rdk_path	40	45,036	896	7.38	0.62
topological_torsion	40	23,653	421	3.47	0.56
ecfp6	37	43,299	731	6.02	0.53
layered	40	43,642	674	5.55	0.48
morgan	38	40,827	398	3.28	0.31

Table S4: Model coverage: datasets attempted and datasets yielding a valid metric, per model.

model	datasets_attempted	datasets_valid
Uni-Mol V2 (164m)	4	0
Uni-Mol V2 (310m)	8	0
Chemprop v2 (CMPNN, ensemble=3)	44	6
Chemprop v2 (D-MPNN + RDKit2D, ensemble=3)	44	6
Chemprop v2 (D-MPNN, ensemble=3)	44	6
Chemprop v2 (D-MPNN + Selected descriptors, ensemble=3)	44	6
Chemprop v2 (AttentiveFP, ensemble=3)	44	6
Uni-Mol V2 (84m)	44	17
ElasticNetCV	22	20
SVR	22	22
SVC	22	22
Voting Classifier (KNN, SVM)	22	22
MapLight CatBoost (Strict Parity)	22	22
Tabular CNN	22	22
LogisticRegression	22	22
Voting Regressor (KNN, SVM)	22	22
HistGradientBoosting	44	44
Extra trees	44	44
Ensemble (Weighted average (inverse train RMSE))	44	44
Ensemble (OOF Stacking (RidgeCV, 5-fold))	44	44
ChemML MLP (PyTorch)	44	44
CatBoost	44	44
CFA (Combinatorial Fusion)	44	44
ChemML MLP (TensorFlow)	44	44
AdaBoost	44	44
Tabular MLP	44	44
MapLight + GNN (CatBoost, Strict Parity)	44	44
Random forest	44	44
Uni-Mol V1	44	44
XGBoost	44	44

Table S5: Run-to-run comparison: the canonical NSF ACCESS Jetstream2 A100 benchmark against the earlier consumer-GPU (RTX 4060) run, analysed identically. "Same split" marks datasets where both runs used the identical held-out partition; the remainder were re-split to scaffold splits in the A100 run and are therefore not directly comparable. Change is relative and metric-direction aware (positive favours the A100 run).

Dataset	Task	Same split	A100 best model	A100 value	RTX 4060 best model	RTX 4060 value	Change (%)
tdc_ames	classification	True	Ensemble (OOF Stacking (RidgeCV, 5-fold))	0.876	XGBoost	0.875	0.142
tdc_bbb_martins	classification	True	Random forest	0.925	Random forest	0.932	-0.788
tdc_bioavailability_ma	classification	True	CatBoost	0.777	CatBoost	0.777	0.000
tdc_cyp1a2_veith	classification	True	Ensemble (Weighted average (inverse train RMSE))	0.971	Ensemble (Weighted average (inverse train RMSE))	0.970	0.091
tdc_cyp2c19_veith	classification	True	Ensemble (Weighted average (inverse train RMSE))	0.923	Ensemble (Weighted average (inverse train RMSE))	0.921	0.194
tdc_cyp2c9_substrate_carbonmangels	classification	True	Ensemble (OOF Stacking (RidgeCV, 5-fold))	0.466	Ensemble (OOF Stacking (RidgeCV, 5-fold))	0.482	-3.140
tdc_cyp2c9_veith	classification	True	Ensemble (OOF Stacking (RidgeCV, 5-fold))	0.814	Ensemble (OOF Stacking (RidgeCV, 5-fold))	0.810	0.481
tdc_cyp2d6_substrate_carbonmangels	classification	True	Uni-Mol V1	0.652	Ensemble (Weighted average (inverse train RMSE))	0.673	-3.096
tdc_cyp2d6_veith	classification	True	Ensemble (OOF Stacking (RidgeCV, 5-fold))	0.734	XGBoost	0.722	1.565
tdc_cyp3a4_substrate_carbonmangels	classification	True	Random forest	0.706	LogisticRegression	0.717	-1.494
tdc_cyp3a4_veith	classification	True	Ensemble (OOF Stacking (RidgeCV, 5-fold))	0.884	Ensemble (OOF Stacking (RidgeCV, 5-fold))	0.890	-0.637
tdc_dili	classification	True	Uni-Mol V1	0.922	Ensemble (OOF Stacking (RidgeCV, 5-fold))	0.915	0.760
tdc_herg	classification	True	AdaBoost	0.848	AdaBoost	0.861	-1.437
tdc_herg_karim	classification	True	Ensemble (Weighted average (inverse train RMSE))	0.905	Ensemble (Weighted average (inverse train RMSE))	0.902	0.355
tdc_hia_hou	classification	True	Ensemble (OOF Stacking (RidgeCV, 5-fold))	0.988	CFA (Combinatorial Fusion)	0.990	-0.187
tdc_pgp_broccatelli	classification	True	Uni-Mol V2 (84m)	0.933	Ensemble (Weighted average (inverse train RMSE))	0.929	0.423
chemml_cep_homo	regression	True	Ensemble (Weighted average (inverse train RMSE))	0.099	TabPFNRegressor	0.086	-15.648
chemml_organic_density	regression	True	Chemprop v2 (D-MPNN + RDKit2D, ensemble=3)	0.005	TabPFNRegressor	0.005	-1.200
esol_delaney	regression	True	Ensemble (Weighted average (inverse train RMSE))	0.617	Ensemble (Weighted average (inverse train RMSE))	0.592	-4.220
freesolv_sampl	regression	True	ChemML MLP (PyTorch)	0.993	Chemprop v2 (AttentiveFP, ensemble=1)	1.080	8.015
lipophilicity	regression	True	Ensemble (Weighted average (inverse train RMSE))	0.589	Ensemble (Weighted average (inverse train RMSE))	0.582	-1.272

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Dataset	Task	Same split	A100 best model	A100 value	RTX 4060 best model	RTX 4060 value	Change (%)
poduam_pod_nc_std	regression	True	Random forest	0.720	Ensemble (Weighted average (inverse train RMSE))	0.699	-3.062
poduam_pod_rd_std	regression	True	Random forest	0.572	XGBoost	0.551	-3.896
polaris_adme_fang_hppb_1	regression	True	MapLight + GNN (CatBoost, Strict Parity)	0.448	MapLight + GNN (CatBoost, Strict Parity)	0.449	0.188
polaris_adme_fang_perm_1	regression	True	Uni-Mol V1	0.399	Uni-Mol V1	0.399	-0.040
polaris_adme_fang_rclint_1	regression	True	Uni-Mol V1	0.522	Uni-Mol V1	0.512	-1.976
polaris_adme_fang_rppb_1	regression	True	MapLight + GNN (CatBoost, Strict Parity)	0.493	MapLight + GNN (CatBoost, Strict Parity)	0.494	0.170
polaris_adme_fang_solu_1	regression	True	Uni-Mol V2 (84m)	0.551	Uni-Mol V1	0.575	4.202
tdc_caco2_wang	regression	True	Ensemble (Weighted average (inverse train RMSE))	0.342	Ensemble (Weighted average (inverse train RMSE))	0.336	-1.531
tdc_clearance_hepatocyte_az	regression	True	Ensemble (Weighted average (inverse train RMSE))	43.841	MapLight + GNN (CatBoost, Strict Parity)	43.976	0.308
tdc_clearance_microsome_az	regression	True	Uni-Mol V2 (84m)	33.507	Uni-Mol V1	35.888	6.634
tdc_half_life_obach	regression	True	Ensemble (Weighted average (inverse train RMSE))	18.541	Uni-Mol V1	17.222	-7.656
tdc_ld50_zhu	regression	True	ChemML MLP (TensorFlow)	0.828	CFA (Combinatorial Fusion)	0.834	0.687
tdc_lipophilicity_astrazeneca	regression	True	Uni-Mol V1	0.617	Uni-Mol V1	0.587	-5.135
tdc_ppbr_az	regression	True	MapLight + GNN (CatBoost, Strict Parity)	11.356	Ensemble (Weighted average (inverse train RMSE))	11.014	-3.108
tdc_solubility_aqsolddb	regression	True	Uni-Mol V1	1.007	Uni-Mol V1	0.989	-1.796
tdc_vdss_lombardo	regression	True	MapLight + GNN (CatBoost, Strict Parity)	4.670	MapLight + GNN (CatBoost, Strict Parity)	4.668	-0.035
tdc_carcinogens_lagunin	classification	False	Chemprop v2 (AttentiveFP, ensemble=3)	0.929	AdaBoost	0.863	7.656
tdc_clintox	classification	False	CFA (Combinatorial Fusion)	0.974	CatBoost	0.949	2.644
tdc_pampa_ncats	classification	False	Uni-Mol V1	0.763	Ensemble (Weighted average (inverse train RMSE))	0.806	-5.258
tdc_skin_reaction	classification	False	Uni-Mol V2 (84m)	0.658	CFA (Combinatorial Fusion)	0.769	-14.470
tdc_tox21	classification	False	AdaBoost	0.556	XGBoost	0.812	-31.520
tdc_toxcast	classification	False	HistGradientBoosting	0.716	CatBoost	0.791	-9.562
tdc_hydrationfreeenergy_freesolv	regression	False	Ensemble (OOF Stacking (RidgeCV, 5-fold))	1.112	CFA (Combinatorial Fusion)	0.556	-99.848